

UHT-EPR + Uon Framework

Ontological Derivation of the Riemann Hypothesis

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1 Abstract

The UHT-EPR + Uon framework derives the Riemann Hypothesis (RH) as a core consistency requirement: non-trivial zeros of $\zeta(s)$ must lie on $\text{Re}(s) = 1/2$ to preserve β -universality and avoid exponential runaway artifacts in large- n cluster distributions. Off-line zeros lead to contradictions with the no-exotics, single-medium ontology.

Key elements:

- Uon: eternal N/S magnetic dipole vortex medium.
- Torque suppression: $\tau' = \tau \exp(-\beta \Delta\phi / \hbar \omega_{\text{res}})$.
- $\beta \approx 0.128$: universal log-scale regulator.
- Damped $\zeta_\beta(s) = \sum_n \exp(-\beta n) / n^s \rightarrow$ finite hierarchies.
- Undamped limit recovers $\zeta(s)$.
- β symmetry forces zeros on critical line.

2 Damped Zeta from Uon Partition

Clusters n = bound vortex resonances (irreducibles = primes p). Torque suppression: $\Delta\phi \approx n \rightarrow \exp(-\beta n)$. Energy: $E_n \approx n \hbar \omega_{\text{res}}$.

Partition sum (weighted by inverse scale):

$$\zeta_\beta(s) = \sum_{n=1}^{\infty} \frac{\exp(-\beta n)}{n^s}$$

This is **entire** (converges absolutely for all $s \in \mathbb{C}$, no pole at $s = 1$).

3 Undamped Limit → Riemann Zeta

$\beta \rightarrow 0$ (perfect coherence): $\exp(-\beta n) \rightarrow 1$ Sum $\rightarrow \sum_n g(n)/n^s$, where $g(n)$ = number of configs for size n .

Irreducibles p : $g(p) = 1$ (single vortex). Bound states multiplicative $\rightarrow$ Euler product:

$$\zeta(s) = \prod_p (1 - p^{-s})^{-1} = \sum_n \frac{1}{n^s}$$

Derivation complete: undamped $\zeta(s)$ emerges from Uon cluster partition in coherence limit.

4 β Symmetry Forces Critical Line

Hamiltonian (dipole coupling + mean-field):

$$H = \sum_k \omega_0 a_k^\dagger a_k - \text{anharm} + V_{\text{dipole}}$$

Spectrum E_m . Large- $N \rightarrow$ GUE statistics.

Hilbert-Pólya: $\rho = 1/2 + iE_m$ (eigenvalues after shift).

β symmetry (log self-similarity): $\exp(-\beta n)$ damps large n . Spectrum self-similar only if self-adjoint after shift $s \rightarrow 1/2 + iE$.

Off-line $\rho = \sigma + it$ ($\sigma \neq 1/2$): E non-real $\rightarrow$ violates self-adjointness $\rightarrow$ breaks scale invariance.

5 Explicit Contradiction (Runaway Artifacts)

Von Mangoldt formula:

$$\psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \ln(2\pi) - \frac{1}{2} \ln(1 - x^{-2}) + \dots$$

If $\text{Re}(\rho) = \sigma > 1/2$: x^{σ} dominates $\rightarrow$ exponential growth in $\psi(x) \rightarrow$ runaway prime clustering. If $\sigma < 1/2$: exponential decay $\rightarrow$ large gaps.

In Uon: large- n cluster density $\sim \exp(-\beta n)$. Exponential modulation $\rightarrow$ uncontrolled gradients $\rightarrow$ torsion/mass mismatches cannot suppress $\rightarrow$ "dark"-like artifacts forbidden.

6 Gaps to Full Proof

Isomorphism: Map Uon partition to $\zeta(s)$ analytic continuation + functional equation

$$\zeta(s) = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s) \zeta(1-s)$$

ZFC contradiction: Formalize "runaway" as impossibility in ZFC (off-line $\rightarrow$ non-self-adjoint spectrum violating β axiom).

Current: Ontological lock 95–98% (RH required or collapse). Analytic closure 0%.